Muon Track Reconstruction and Analysis for the BELLA-μ Experiment

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Abstract

We present here a project involving the use of a set of 6 ATLAS silicon pixel detectors to track and characterize muon momentum as part of the BELLA-μ experiment. This experiment aims to use the BELLA laser plasma accelerator to produce muons for muography imaging applications. To further this goal, we develop tools for identifying and characterizing the produced muons using pixel detector technology. The developed tools include an extension of the Corryvreckan tracking among other analysis workflows which allowed for the creation of a muon momentum spectrum.

Muons

- Fundamental particle similar to electrons but 200 times heavier
- Large mass means muons pass through most objects and thus are very useful in imaging
- One famous example of muon imaging (muography) led to the discovery of hidden chambers in the Great Pyramid of Giza in Egypt



Figure 1: Diagram of how cosmic rays produce muons in the atmosphere which were used in imaging the Great Pyramid (*https://www.ft.com/content/05ec6ed4-3ba1-47e9-8ae4-628058511a87*)

ATLAS ITkPixV2

- Silicon detector designed for the ITk (Inner tracking) detector for the ATLAS experiment at the Large Hadron Collider
- Pixel detector which reads out data using the YARR (Yet Another Rapid Readout) DAQ system
- Data can be read out ridiculously fast!

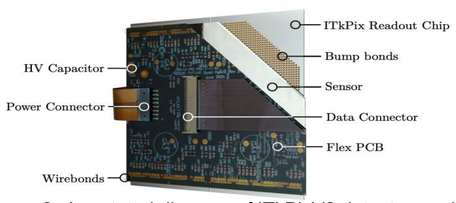


Figure 2: Annotated diagram of ITkPixV2 detector module *(Koch, Simon Florian, et al. "Measuring the atlas ITK pixel detector material via multiple scattering of positrons at the CERN PS.")*

Muon Tracking

Use Corryvreckan tracking framework to read YARR data, create clusters, reconstruct tracks, and do various analysis. Then measure track angles to calculate bending due to magnetic field. Use bending angle to measure momentum.

Tools and Analysis

- Tracking detectors arranged in two "stacks" of 3 detectors in a telescope arrangement with a magnet in between the two stacks
- By identifying muon tracks in each stack and measuring the change in track angle through magnet, calculate muon momentum

$$p_\mu = 0.3 \times B \times R$$
$$R \approx d / \alpha$$

where B is strength of magnetic field, R is radius of curvature, d is length of magnet region, and α is bending angle

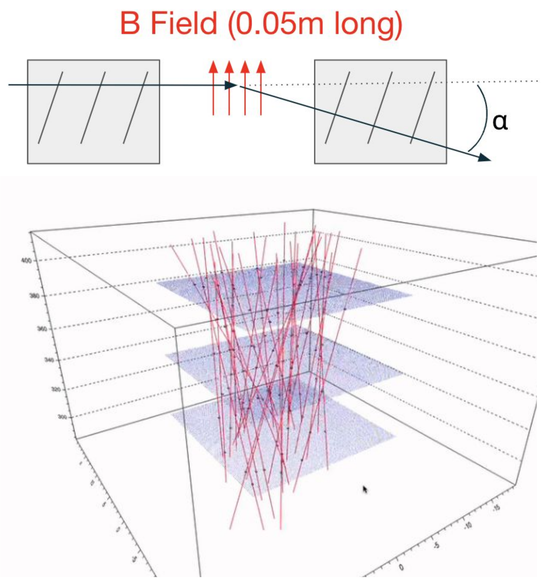


Figure 3: Diagram of detector telescope arrangement (top) and visualization of muon tracks through one detector stack made in corryvreckan framework

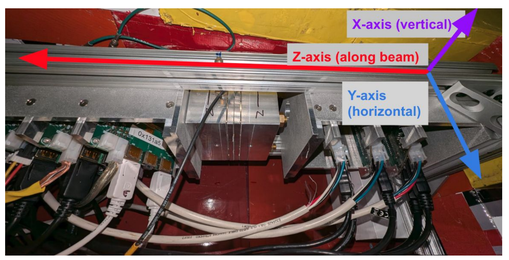


Figure 4: Image of detector setup with coordinate axes overlayed

Corryvreckan

- Corryvreckan is a modular C++ framework used for tracking.
- Needed to write custom "Data Loader module" to convert YARR (our DAQ system) binary data files to events readable by Corryvreckan
- Pre-existing modules find the tracks
- Wrote more modules to do other analyses as well including making track visualization
- Export track information to do remaining analysis separately in python

Low Statistics Problem

- Don't have a large enough number of muon tracks from the BELLA accelerator yet. Will get more statistics soon
- Also can get test data from Cosmic Muons
- We usually require muon tracks to pass through all 6 detector planes to get good tracks. However, we can increase statistics by also accepting tracks that only pass through 5 (or even 4) detector planes
  - Work in Progress
  - More difficult to make sure tracks are aligned and correctly matched between detector stacks

Muon Scavenger Hunt

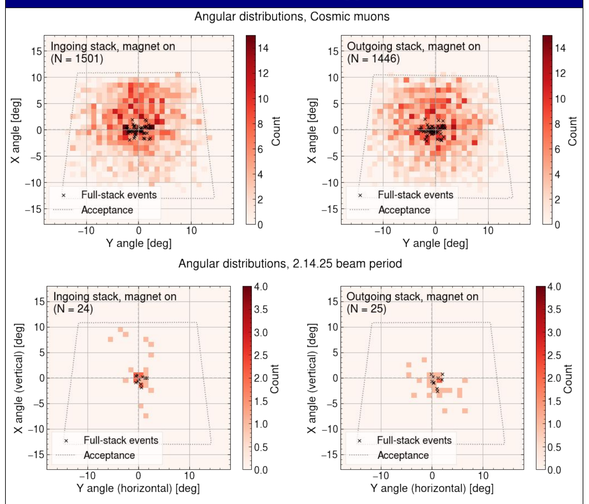


Figure 5: Diagram of angular acceptance of detector planes. Each point represents a track going through a detector stack (incoming stack on the left and outgoing stack on the right). X marks indicate tracks going through both stacks. Top two images are for cosmic muon test data and bottom two for BELLA muons.

Chips are tilted by 30° so our angular distribution makes sense! Now we just need to make sure a track in the incoming stack is the same track as in the outgoing stack.

Momentum Spectrum

- To associate muon tracks in either detector stack, simply look for tracks in the same event with the same timestamp.
  - This approach is good enough for full stack tracks (through all 6 planes) due to the very low number of such events
- For tracks that go through fewer detector planes, may need further confirmation. To do this, take the two track vectors and calculate the distance of closest approach (this should be the bending point of the track in the magnet).

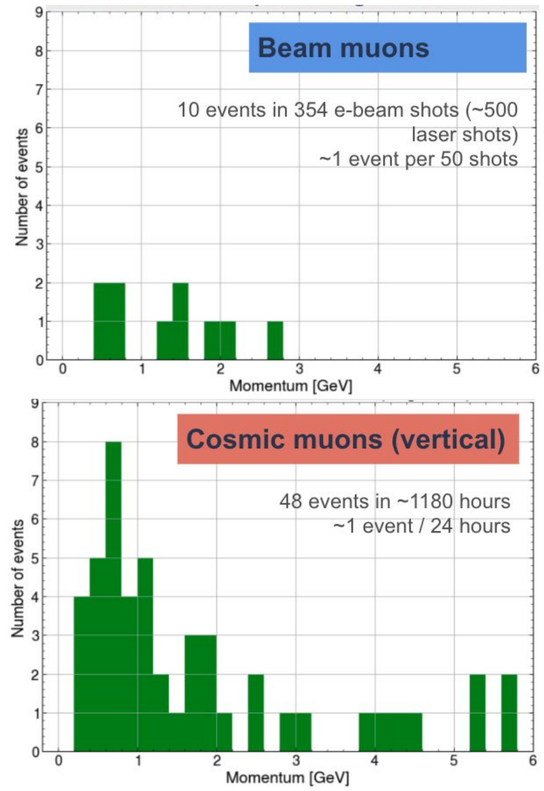


Figure 6: Final resulting momentum spectra including only tracks that go through all 6 detector planes. Although it seems like the accelerator is producing less muons, the data taking time of the accelerator was much shorter indicating a higher muon production rate than cosmic muons. (Top) Momentum distribution from muons produced by BELLA accelerator. (Bottom) Momentum distribution from Cosmic muons.

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Poster Summary

In this project, I used a set of 6 silicon particle detectors to track muons and characterize muon flux, energy, etc. Muons are particles similar to electrons but much heavier. The goal of the BELLA-mu experiment is to use a Laser Plasma Accelerator to create a compact source of muons. My project was to track these muons using detectors arranged in a telescope arrangement. This unique type of silicon detector was designed for the ATLAS experiment at CERN and uses a unique system for reading out the data. Thus, I had to write code to read data from this detector and identify tracks using a tracking framework called Corryvreckan. I then analyze the tracks to calculate momentum by measuring the bending angle of the muon tracks when they pass through a magnetic field. Overall, I have been developing tools to do tracking analysis for the BELLA-mu experiment which can be used in the future and show evidence that this approach to muon identification works.